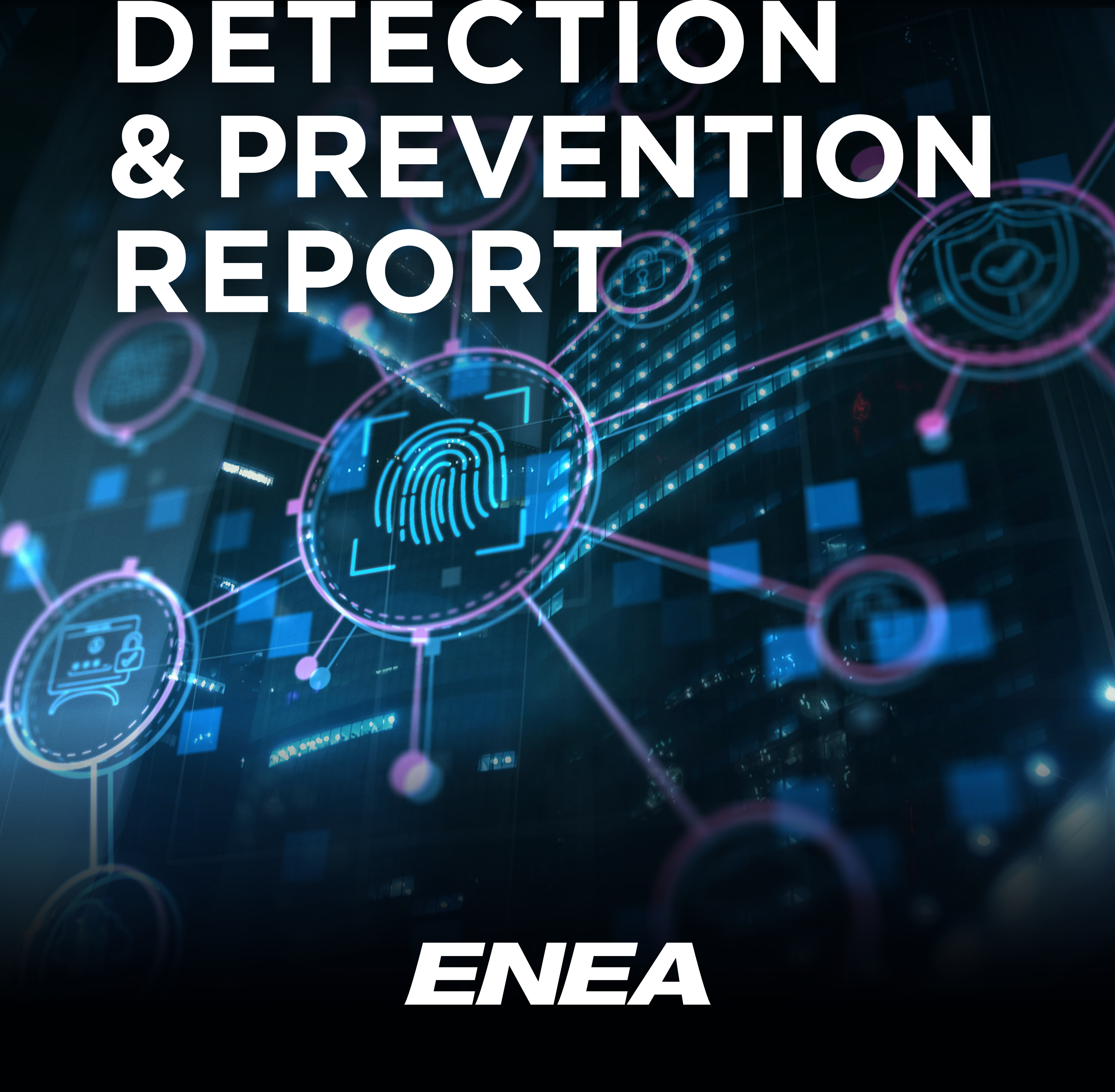
2022

Cybersecurity INSIDERS

INTRUSION DETECTION & PREVENTION REPORT



ENEA

EXECUTIVE SUMMARY

Intrusion Detection & Prevention Systems (IDS/IPS) have long provided a critical last line of defense against sophisticated cybersecurity attacks that have penetrated endpoint and perimeter defenses. They also offer an important frontline defense in emerging networks that do not really have fixed perimeters, and which often connect new kinds of devices that cannot support conventional embedded endpoint security software.

However, conventional IDS/IPS have some weaknesses in usability (like generating high volumes of false positive alerts) and in effectiveness (such as a blindness to many protocols and to certain types of advanced threats – especially those using encryption to evade detection). The latter is especially worrying as the use of encryption expands and standards become more rigorous, making it impossible to use common current methods of analyzing traffic without resorting to decryption.

To better understand these concerns, how IDS/IPS is currently being used, and what the future might hold for this important technology, we conducted a comprehensive survey of Cybersecurity Insiders 500,000-member information security community.

For a panel discussion about options and strategies for addressing the needs and concerns raised in this survey, we invite you to watch our webinar [2022 State of IDS/IPS: Adapt or Die.]

Many thanks to [Enea Qosmos] for supporting this important research project, and we hope you find the information shared by respondents useful in strengthening your own cybersecurity posture.

Thanks

Holger Schulze

Holger Schulze
CEO and Founder
Cybersecurity Insiders

Cybersecurity
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KEY FINDINGS

IDS/IPS adoption is widely deployed

IDS/IPS remains one of the most widely deployed cybersecurity technologies. 80% of respondents report having deployed one or more IDS/IPS systems. In addition, while 39% of respondents use standalone systems, IDS/IPS functionality is increasingly being integrated into solutions like Network Detection and Response (NDR) (21%), Extended Detection and Response (XDR) (29%), and Cloud Firewalls (CFW) (26%). This indicates that network-based threat detection remains an essential tool for combatting advanced threats.

IDS/IPS cloud deployments increase

IDS/IPS is still widely deployed on-premise (65%), but significantly, a nearly equal number (64%) now report deploying IDS/IPS in the cloud, including both private (35%) and public (29%) clouds. The category of protocols for which respondents would most like to see improved coverage is business Software as a Service (SaaS) applications. This indicates organizations are assuming much more responsibility for cloud security.

IDS/IPS capabilities are improving

The survey indicates significant improvement has been made in Anomaly Detection (AD) capabilities in IDS/IPS, with respondents citing anomaly detection as their 2nd most relied upon IDS/IPS function (71%). However, the demand for more accurate and actionable IDS/IPS alerts persists, with respondents specifically calling out a need for more definitive attack declarations (45%) and automatic alert triage (42%).

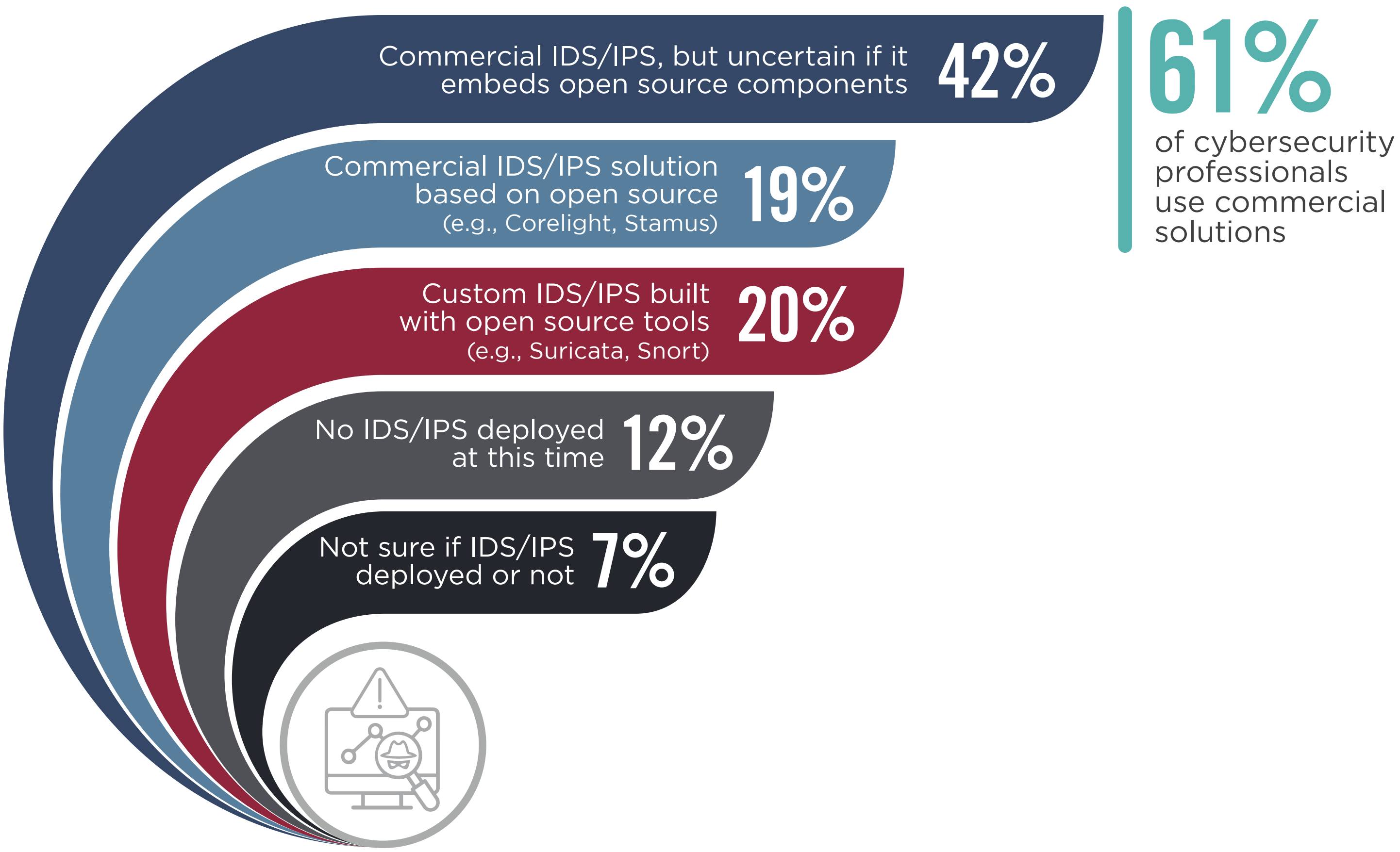
Future prognosis is uncertain

While IDS/IPS use seems rock solid year after year, responses indicate that many legacy IDS/IPS are ill-equipped to meet the future challenge of fully encrypted environments. Nearly half of all respondents also say they are unsure whether open source IDS/IPS software like Suricata and Snort will still be playing an important role in threat detection in the future. This could reflect concerns over specific issues like encryption, or that techniques like Machine Learning/Artificial Intelligence (ML/AI) will replace rather than complement IDS/IPS. It could also simply reflect the fact that a large percentage of respondents are unaware that their commercial IDS/IPS solution embeds open source components. This obfuscation is likely to increase as IDS/IPS is embedded in umbrella solutions like XDR and Cloud Firewalls.

IDS/IPS TYPES

IDS/IPS remains one of the most widely deployed cybersecurity technologies. At least 80% of respondents report their organization has deployed one or more IDS/IPS platforms. Most respondents use commercial IDS/IPS (61%), though only 19% know for sure whether their commercial IDS/IPS embeds open source (in fact, most do). Twenty percent of respondents use open source to develop custom versions.

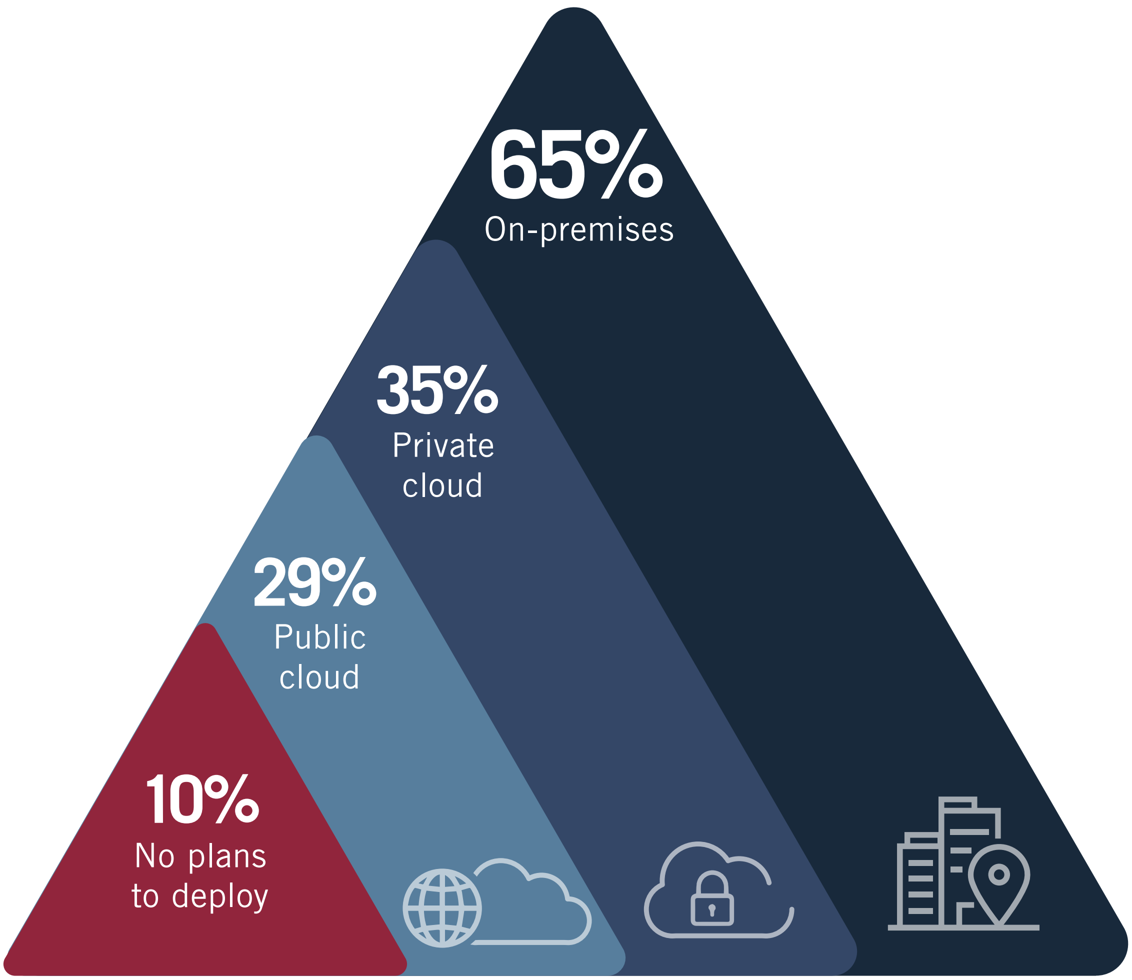
What type of IDS/IPS do you use?



IDS/IPS DEPLOYMENT

IDS/IPS deployments are evolving in step with changes in network environments, particularly the migration of IT assets and systems to the cloud. While IDS/IPS is still widely deployed on-premise (65%), a nearly equal number (64%) now use IDS/IPS in the cloud, including 29% in public and 35% in private clouds. This indicates organizations are becoming more proactive about assuming responsibility for cloud security.

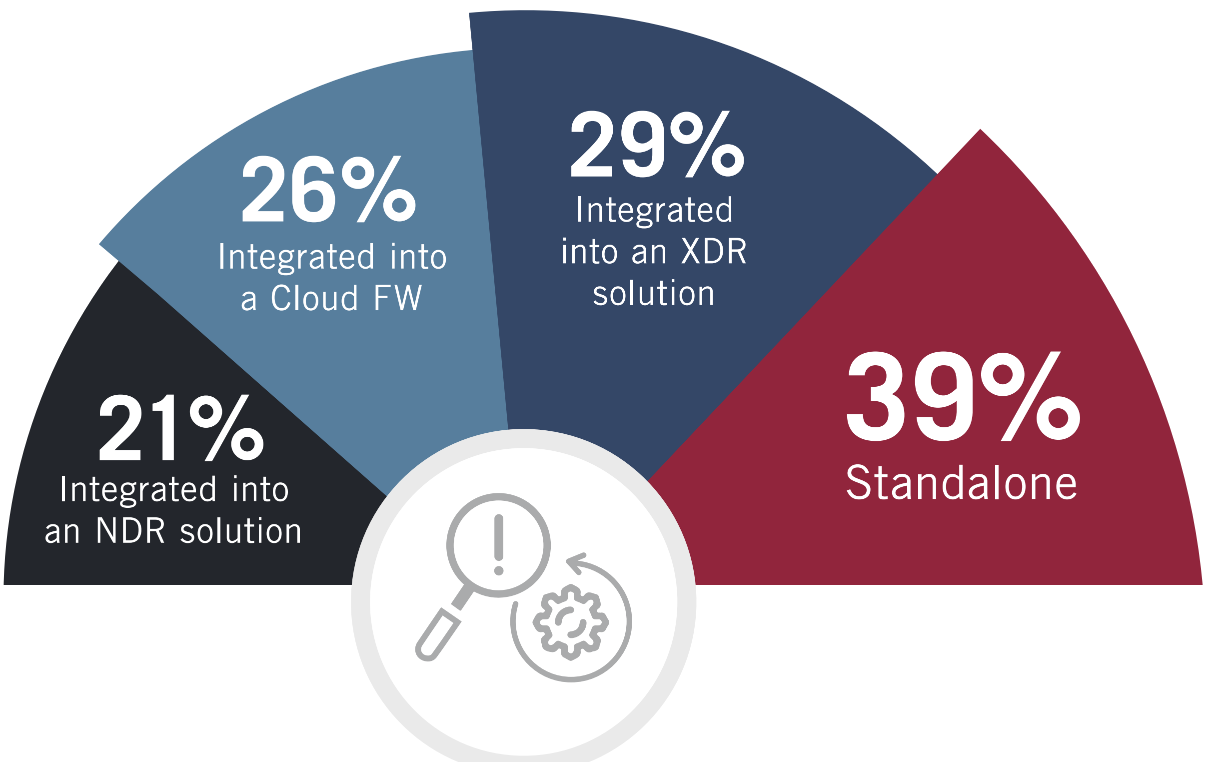
Where have you deployed (or plan to deploy) IDS/IPS? (Select all that apply)



STANDALONE VS. INTEGRATED SOLUTIONS

While 39% of respondents still use IDS/IPS in standalone form, they are increasingly being integrated into more comprehensive threat management solutions. The most popular integration is with Extended Detection & Response (XDR) solutions (29%), which combine network- and endpoint-based threat detection and response to combat advanced attacks. IDS/IPS is also increasingly used to enhance the capabilities of Cloud Firewall (CFW), specifically the use of integrated DPI and IDS/IPS to transform simple cloud firewalls into Next Generation Firewalls (NGFW).

If you use IDS/IPS, is it standalone or integrated? (Select all that apply)

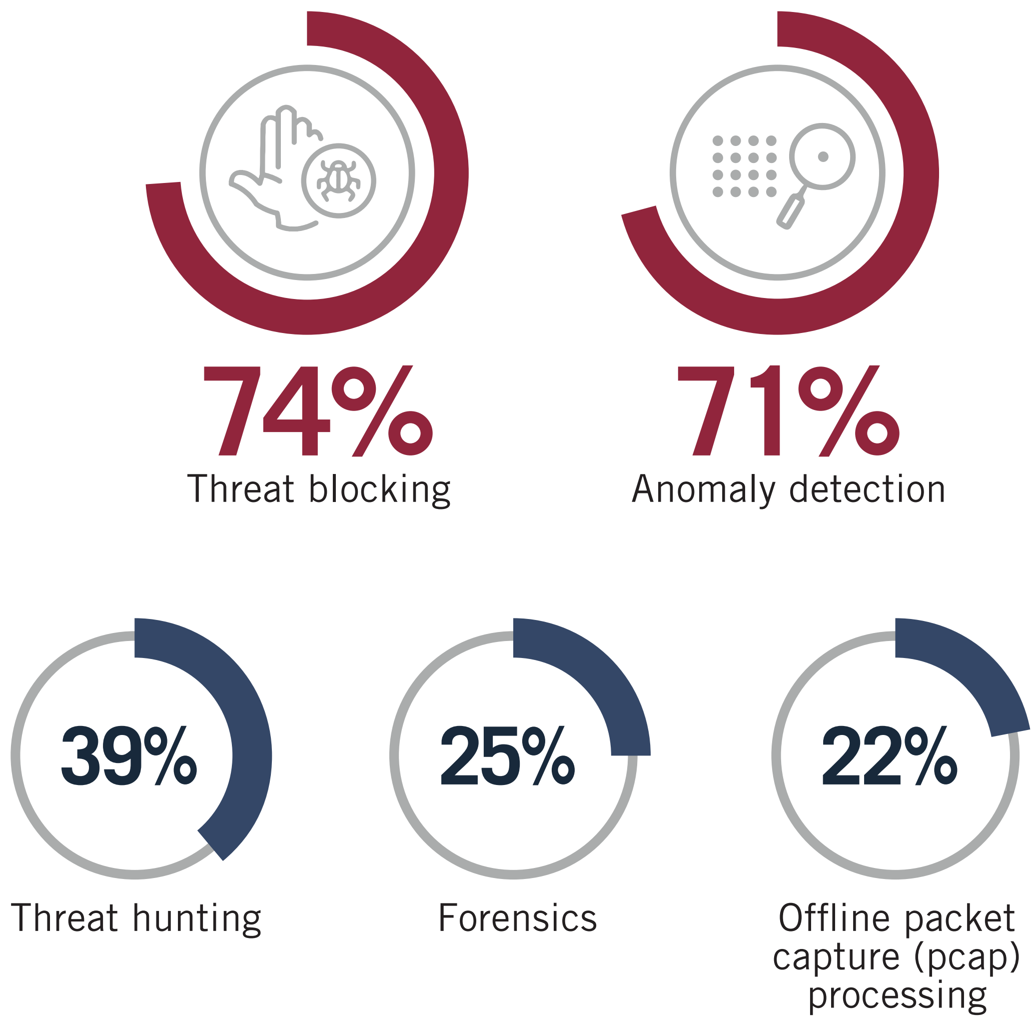


Other 14%

POPULAR IDS/IPS FUNCTIONS

Three quarters of respondents report the most-used IDS/IPS functions are threat blocking (74%) and anomaly detection (71%). The prominent second place showing for anomaly detection (AD) indicates that this functionality within IDS/IPS is rapidly maturing; in last year's survey on Network Detection & Response (NDR), respondents cited anomaly detection (74%) as the primary advantage of NDR over standalone IDS/IPS, which was considered weak on AD.

What type of functions do you use IDS/IPS for? (Select all that apply)

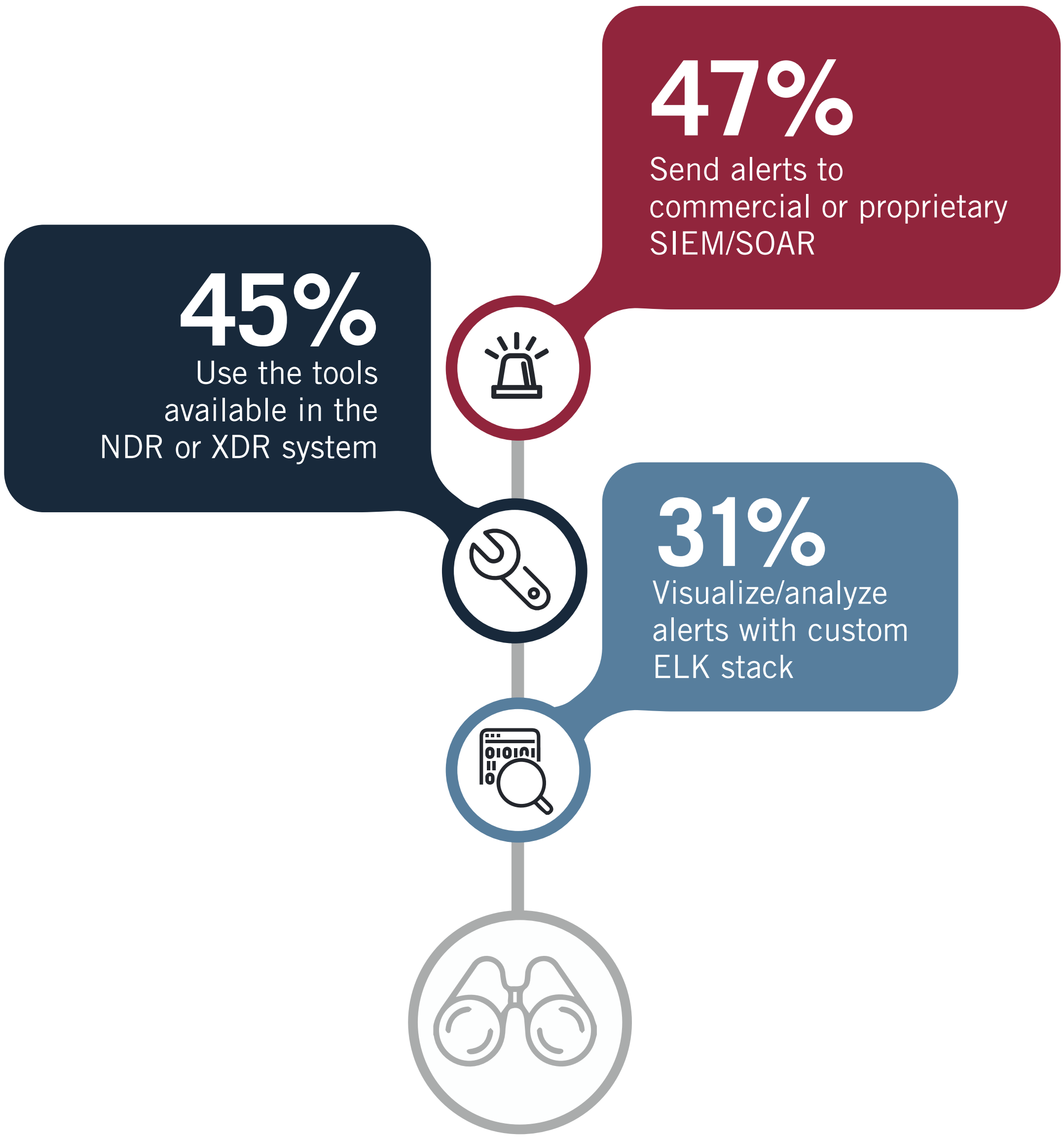


Other 7%

ALERT ANALYSIS

Nearly half of cybersecurity professionals (47%) view and analyze IDS/IPS alerts within umbrella Security Information & Event Management (SIEM) or Security Orchestration, Automation & Response (SOAR) platforms. A nearly equal number use more focused XDR or NDR (45%) for alert analysis and management. Despite the broad use of these standard platforms, 31% of cybersecurity professionals still develop their own visualization solutions, which might be a strategy for addressing some of the reported usability issues with standard IDS/IPS alerting.

How do you visualize and analyze IDS/IPS alerts? (Select all that apply)

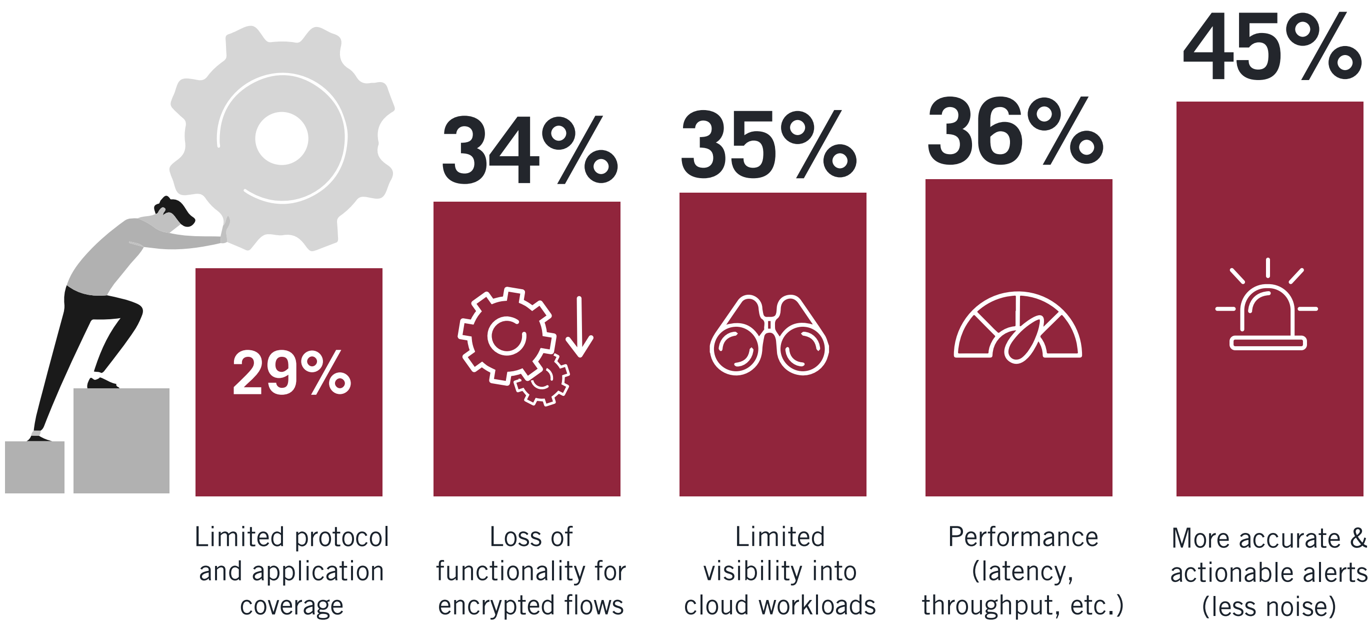


Other 11%

BIGGEST CHALLENGES

For 45% of cybersecurity professionals, inaccurate, 'noisy' alerts are the biggest headache with IDS/IPS solutions. This could help explain why people continue to develop their own analysis and visualization tools for IDS/IPS alerts even when using commercial systems. The other four primary concerns carry nearly equal weight. All challenges noted, except performance challenges, are associated with network traffic visibility issues which can be improved with strategies like DPI integration.

What are your greatest challenges with your IDS/IPS? (Select all that apply)



Other 10%

ENCRYPTED TRAFFIC CHALLENGE

Many IDS/IPS solutions are not prepared for the challenge of fully encrypted environments. Our survey reveals that the most widely used strategy (38%) for detecting threats in encrypted flows is to analyze the minimal data that remains clear in legacy encryption standards only. At least a quarter however are adopting encrypted traffic classification (ETC) technology (26%), while an almost equal number use proxy servers to decrypt and inspect traffic (25%). Using a threat intelligence database is another popular workaround (28%), though this is often integrated into ETC solutions.

How does your IDS/IPS handle threat detection with encrypted flows? (Select all that apply)

Analyze whatever information remains unencrypted in an encrypted flow (e.g., handshake data in non-TLS 1.3 environment) **38%**

Enrich with threat intelligence database **28%**

Extend IDS/IPS with an Encrypted Traffic Classification (ETC) component (e.g., ETC-enabled DPI engine) **26%**

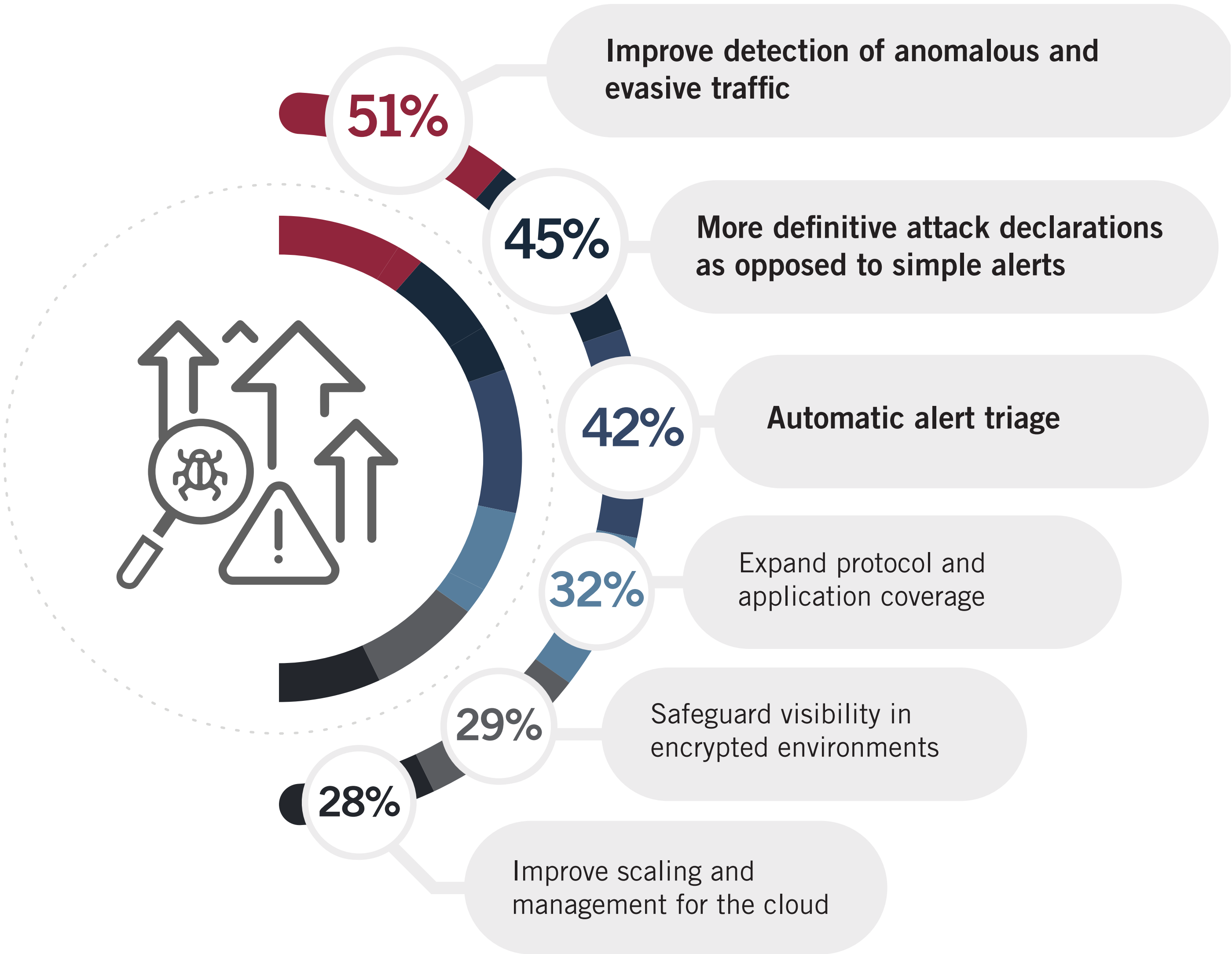
Use MITM/Proxy to decrypt traffic, then inspect **25%**

Not sure/other 25%

NEEDED IMPROVEMENTS

Strengthening anomaly detection tops the list of improvements cybersecurity professionals would most like to see in IDS/IPS technology (51%). The next two desired enhancements, more definitive attack declarations (45%) and automatic alert triage (42%), underscore respondent complaints about noisy alerting. The desire for expanded protocol coverage and preservation of visibility in encrypted environments points to the need to integrate IDS/IPS with advanced DPI.

What would you most like to improve about your IDS/IPS? (Select all that apply)

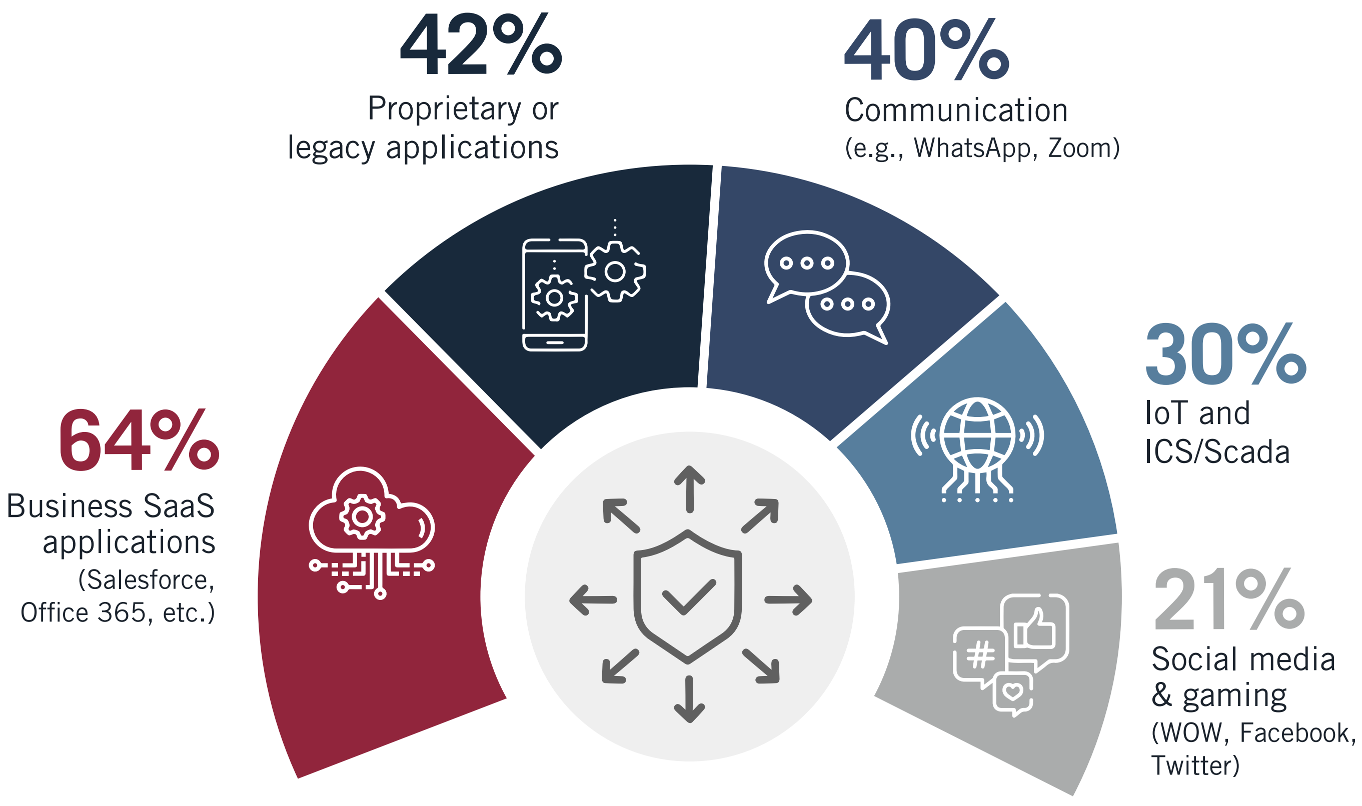


Other 6%

APPLICATION COVERAGE

Cybersecurity professionals' desires related to improvements in IDS/IPS protocol coverage reflect the growing reliance on cloud services in the modern enterprise. When asked in which application categories they would most like to see improved coverage, the number one response was business SaaS apps (64%). Communications applications, which are cloud-based today, come in at third place (40%), behind expanding coverage to proprietary or legacy applications (42%), which in the case of new custom applications are almost all developed on cloud infrastructure. Nearly one third would also like their IDS/IPS to do a better job with Internet of Things (IoT) and Operational Technology (OT) coverage (30%), reflecting the interrelated trends of cloudification and IT/OT convergence.

If you could expand IDS/IPS protocol and application coverage, which application categories would you choose? (Select all that apply)

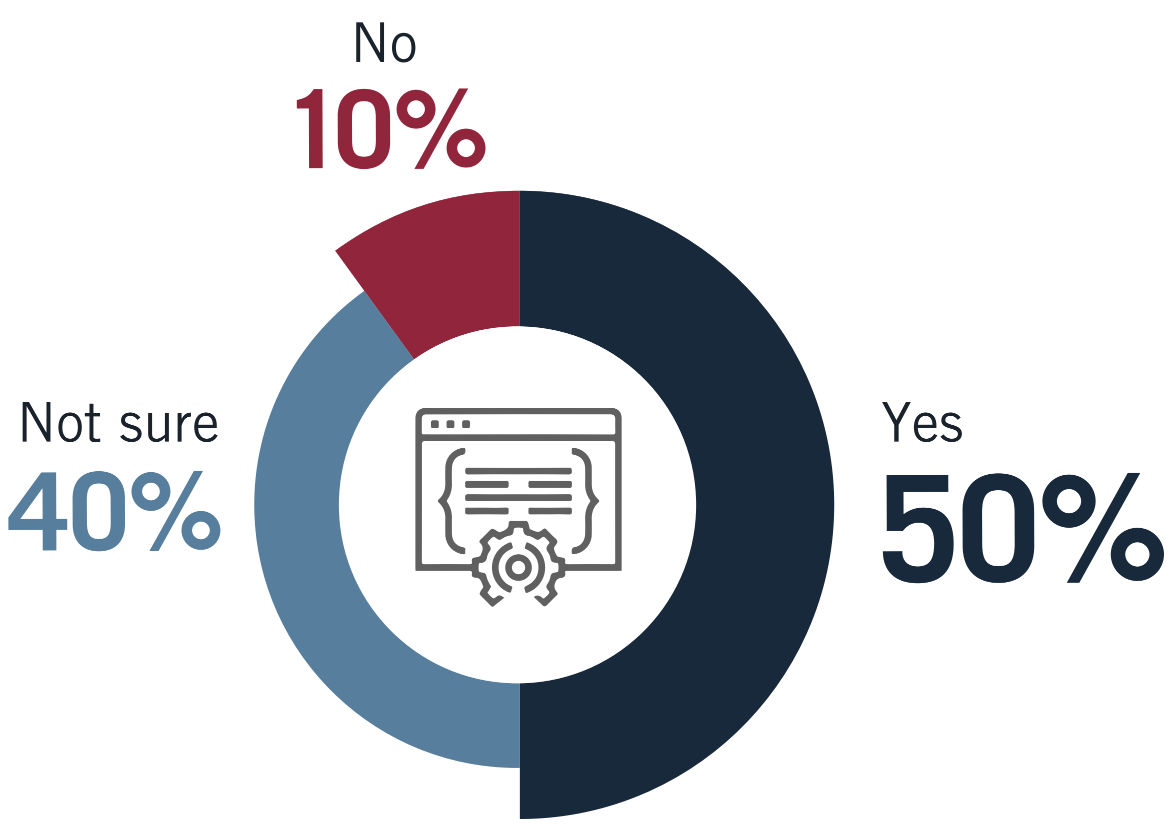


Other 6%

FUTURE IMPACT OF OPEN-SOURCE SOFTWARE

Despite a broad general reliance on IDS/IPS today, only half of respondents are fully confident that open source IDS/IPS will still be playing an important role in threat detection five years from now. Forty percent are unsure, while 10% think it will not. There are multiple possible explanations for this relatively weak level of confidence. One possibility is the low-level awareness of how widespread the use of open source software is in commercial IDS/IPS. Another is perhaps a belief that specific challenges like encryption will limit the effectiveness of all IDS/IPS, or that emerging techniques like machine learning-based threat detection will replace (rather than enhance) IDS/IPS. Or, it may simply reflect an assumption that standalone IDS/IPS (including open source systems like Suricata and Snort) will cease to exist outside encompassing systems like next generation Cloud FWs, XDR and SIEM/SOAR.

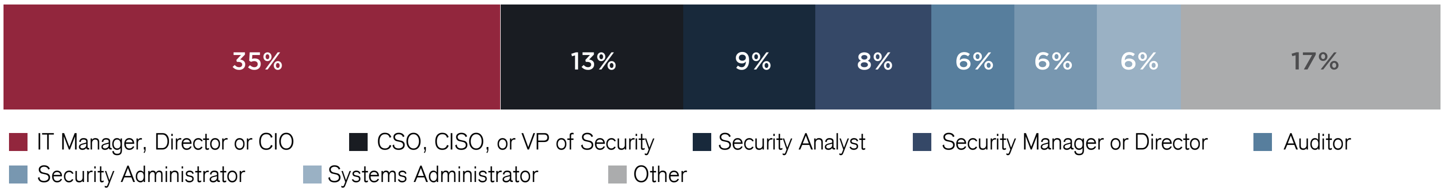
Do you think open source software like Suricata and Snort will still be playing an important role in threat detection five years from now?



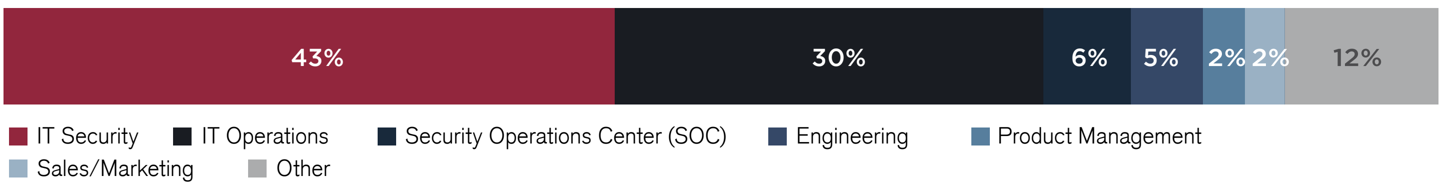
METHODOLOGY & DEMOGRAPHICS

This report is based on the results of a comprehensive online survey of 388 cybersecurity professionals, to gain more insight into the latest trends, key challenges, and solutions for IDS/IPS. The respondents range from technical executives to managers and IT security practitioners, representing a balanced cross-section of organizations of varying sizes across multiple industries.

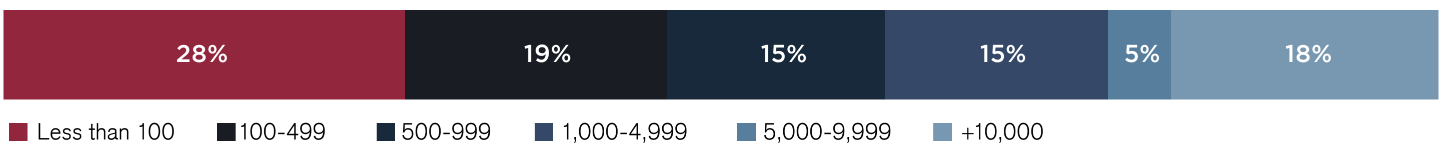
PRIMARY ROLE



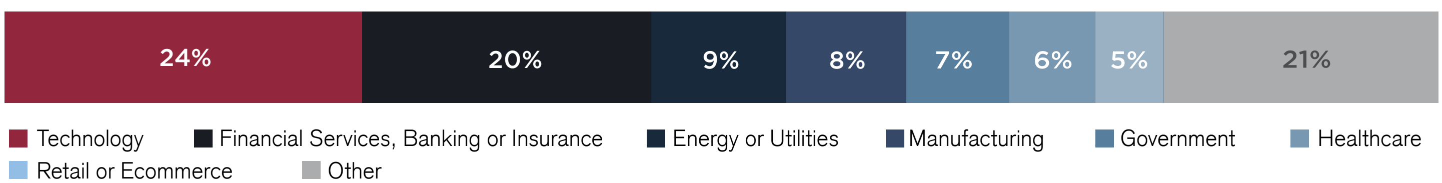
DEPARTMENT



COMPANY SIZE



INDUSTRY



About Enea

Enea is a world-leading specialist in software for telecom and cybersecurity. The company's cloud-native solutions connect, optimize, and secure services for mobile subscribers, enterprises, and the Internet of Things. More than 100 communication service providers and 4.5 billion people rely on Enea technologies every day.

Enea's embedded DPI-based traffic intelligence software classifies traffic in real-time and provides granular information about network activities. Enea's Qosmos technology is trusted by 75% of cybersecurity and networking vendors who embed commercial DPI technology in their solutions. Typical uses cases include SD-WAN, SASE, SSE, ZTNA, NGFW, SWG, XDR.

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